Arya Mahardika

Machine Learning Researcher · Indonesian NLP

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SKILLS

Research
Language modelling, Machine translation, Speech recognition, Evaluation

Engineering
Distillation, Data pipelines, MLOps

Tools
Python, PyTorch, JAX, Hugging Face, Kubernetes

AWARDS

Best Paper Award — 2025
Conference on Low-Resource Language Technology

Young Researcher Grant — 2024
Yayasan Riset Bahasa Nusantara · IDR 600 million

Master's Scholarship — 2016
LPDP

LANGUAGES

Indonesian
Native

Balinese
Native

English
Fluent (IELTS 7.5)

Javanese
Conversational

Summary

Applied scientist building language models that work for Indonesia's regional languages, not just Indonesian. Nine years in NLP research and production, from low-resource machine translation to speech recognition for Javanese and Sundanese. My models serve 4 million requests a day, and my open datasets are used by more than 30 research groups.

Experience

Lentera Nalar AI Lab — Jakarta
Senior Research Scientist, NLP — Jan 2023 – Present

- Lead a team of five training a 7-billion-parameter model for Indonesian and 12 regional languages; it beats open baselines by 9 points on our benchmark.
- Released NusaEval, a 40-task benchmark now used by more than 30 universities and labs.

Suarakata — Bandung
Applied Scientist, Speech — Mar 2020 – Dec 2022

- Built Javanese and Sundanese speech recognition for a call-centre product; word error rate fell from 31% to 14%.
- Designed the pipeline that collected 6,000 hours of speech from 1,900 paid volunteer speakers.

Institut Teknologi Bandung — Bandung
Research Assistant, Computational Linguistics — Sep 2017 – Feb 2020

- Built a 2-million-sentence Indonesian–Javanese parallel corpus, released under an open licence.

Publications

NusaEval: a benchmark for language models in Indonesian and twelve regional languages — 2025
Conference on Low-Resource Language Technology 2025
First author, with S. Hutapea, N. Rahim and M. Tan

Cheap tokens, fair models: vocabulary design for Austronesian languages — 2024
Journal of Computational Linguistics in Asia 8(1)

Crowd-sourced speech corpora for Javanese and Sundanese ASR — 2022
Workshop on Speech for Under-Resourced Languages

Education

University of Edinburgh — Master of Science · With distinction
Artificial Intelligence — Edinburgh, United Kingdom · 2016 – 2017

Institut Teknologi Bandung — Bachelor of Engineering · GPA 3.71
Informatics — Bandung · 2011 – 2015